1. Project Overview

hbd_chatbot is an intelligent, multi-intent chatbot platform designed to serve both business owners and end-users through a unified conversational interface. The system leverages a Large Language Model (LLM) backend combined with structured database queries and real-time online search capabilities to deliver accurate, context-aware responses.

The platform identifies user intent dynamically and routes requests through specialized processing pipelines — ensuring that business management tasks, data search queries, and general FAQ interactions are all handled with precision and efficiency.

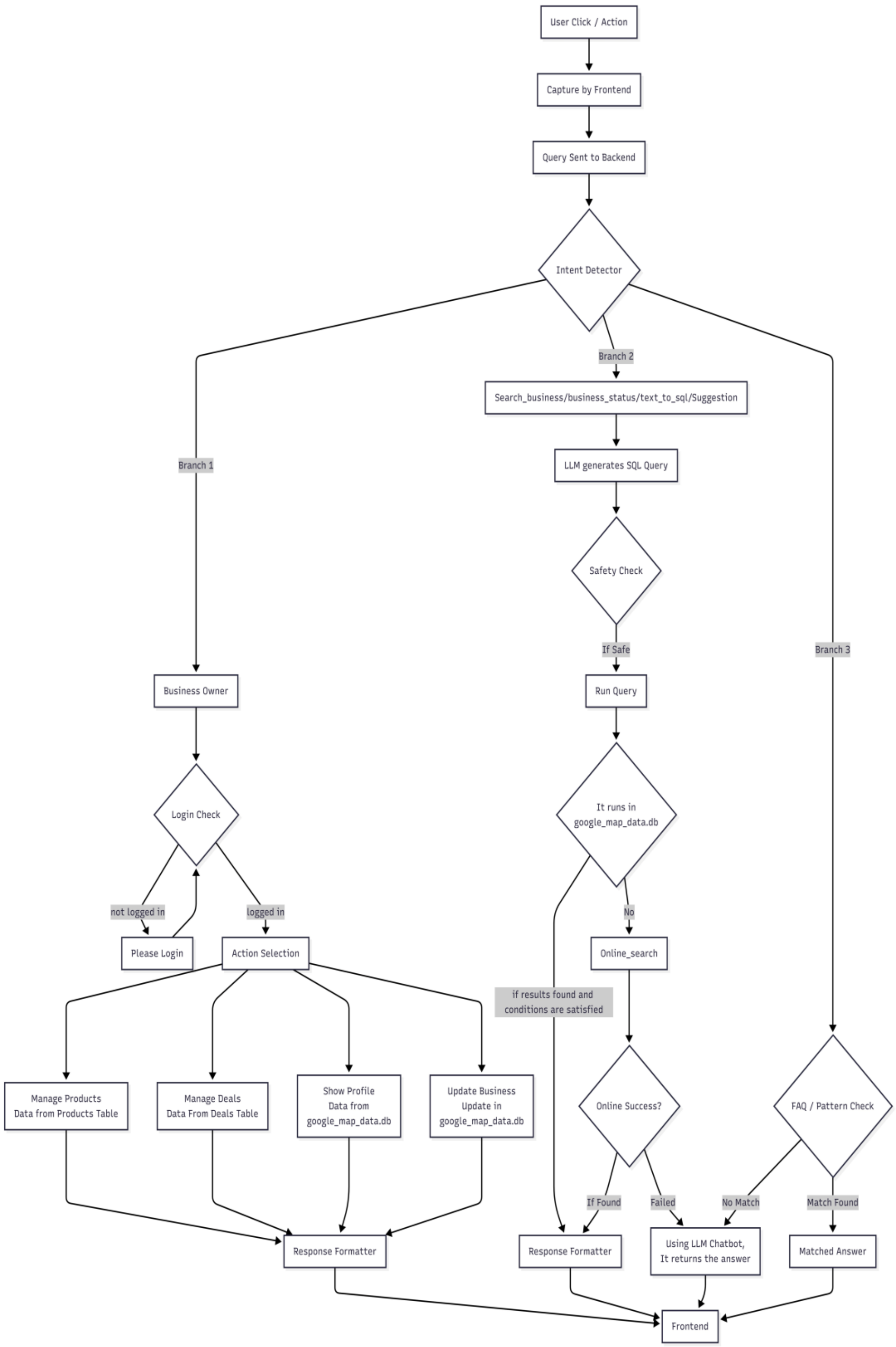
1.1 Key Objectives

- Provide a seamless conversational interface for multiple user intents
- Enable business owners to manage products, deals, and profile information
- Support natural language to SQL translation for intelligent data querying
- Integrate online search as a fallback for real-time data needs
- Ensure all LLM-generated queries pass safety validation before execution
- Deliver consistent, formatted responses to the frontend across all branches

2. System Architecture

The hbd_chatbot system follows a branched routing architecture. All user interactions enter through a single Frontend gateway, are captured and forwarded to the Backend, and are then classified by an Intent Detector. Based on the detected intent, the system routes processing through one of three distinct branches before returning a formatted response to the user.

2.1 Architecture Diagram



3. Request Processing Flow

3.1 Entry Point

- User Click / Action: The user performs an action or sends a message via the frontend interface.
- Capture by Frontend: The frontend captures the raw event/query and packages it for transmission.
- Query Sent to Backend: The packaged query is transmitted to the backend server for processing.
- Intent Detector: A core classification module analyzes the query and routes it to the appropriate processing branch.

3.2 Branch 1 — Business Owner Management

This branch handles all requests related to business owner operations. Once identified, the system performs a Login Check:

- Not Logged In: The user is prompted to log in before proceeding.
- Logged In: The user is directed to an Action Selection menu offering four core capabilities:
- Manage Products — retrieves and manages data from the Products Table.
- Manage Deals — retrieves and manages data from the Deals Table.
- Show Profile — fetches business profile data from google_map_data.db.
- Update Business — updates business information in google_map_data.db.

All outputs from this branch are passed to the Response Formatter before being sent to the Frontend.

3.3 Branch 2 — Search / Text-to-SQL / Suggestion

This branch handles intelligent data search requests. The flow proceeds as follows:

- The LLM generates an SQL query based on the natural language input.
- A Safety Check validates the generated SQL to prevent harmful or destructive queries.
- If safe, the query is executed in google_map_data.db.
- If the result found and conditions are satisfied then result is returned.
- If the data is not found locally (No match in db), an Online Search is performed.
- Online Search Success: Results are passed to the Response Formatter.
- Online Search Failed / No Match: The LLM Chatbot generates a conversational answer as a fallback.

All results are routed through the Response Formatter and returned to the Frontend.

3.4 Branch 3 — FAQ / Pattern Matching

This branch handles common, pre-defined queries through a pattern matching engine:

- FAQ / Pattern Check: The system checks the input If it matches to FAQ patterns.
- Match Found: A pre-defined Matched Answer is returned directly.
- No Match: The query falls through to the LLM Chatbot for a generative response.

All responses converge at the Frontend for final rendering and chat history is stored.